Introduction to Large Language Models
Assignment- 10

TYPE OF QUESTION: MCQ
Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1: [1 mark]

In LoRA, the update to a frozen weight matrix W of shape d x k is parameterised as delta_W = B A, with B of shape d x r and A of shape r x k. For d = k = 4096 and rank r = 8, how many trainable parameters does this adapter introduce for that one matrix?

a. 65,536
b. 32,768
c. 16,777,216
d. 8,192

Correct Answer: a

Explanation:

A holds r x k = 8 x 4096 = 32,768 parameters and B holds d x r = 4096 x 8 = 32,768, giving 65,536 in total.

The frozen matrix itself has 4096 x 4096 = 16,777,216 parameters, which is option (c). The adapter is therefore about 0.39 percent of the size of the layer it adapts, which is the whole point of the method.

QUESTION 2: [1 mark]

Where do prefix tuning and adapters inject their task-specific parameters in a Transformer?

a. Prefix tuning inserts bottleneck feed-forward modules inside each block, while adapters prepend trainable vectors to the input.
b. Prefix tuning prepends trainable key and value vectors at every layer, while adapters insert small bottleneck modules inside the blocks.
c. Both modify only the final output projection, differing in how they initialise it.
d. Both work purely by altering the attention mask.

Correct Answer: b

Explanation:

Prefix tuning learns continuous vectors that act as extra keys and values the existing tokens can attend to, added at the input of every layer, with the backbone frozen. Adapters instead add a small down-projection, non-linearity and up-projection inside each block, usually after the attention or feed-forward sub-layer, wrapped in a residual connection.

Option (a) is the same description with the two methods swapped, which is the trap in this question.

QUESTION 3: [1 mark]

Which of the following statements about extensions of LoRA are correct?

a. QLoRA trains low-rank adapters on top of a base model quantised to 4 bits.
b. DyLoRA trains over a range of ranks so that a rank can be selected afterwards without retraining.
c. AdaLoRA distributes a parameter budget across modules by importance instead of fixing one rank everywhere.

d. LoRA adapters cannot be folded back into the base weights, so they always add inference latency.

Correct Answers: a, b, c

Explanation:

(a) True: QLoRA keeps the frozen base model in 4-bit precision and back-propagates through it into the adapters, which is what allows large models to be fine-tuned on a single GPU.

(b) True: DyLoRA trains the decomposition so that its leading blocks are usable on their own, making the rank a choice at deployment time rather than at training time.

(c) True: AdaLoRA prunes rank where it contributes little and reallocates it where the gradients suggest it matters.

(d) False: because delta_W = B A has the same shape as W, the product can be added into W once training is finished, giving exactly the original inference cost. Keeping it separate is a convenience for swapping tasks, not a requirement.

QUESTION 4: [1 mark]

Why does unstructured magnitude pruning often fail to deliver a wall-clock speedup even at high sparsity?

a. Because it removes entire attention heads and so changes the output dimension.

b. Because the surviving weights are scattered irregularly, and dense kernels cannot exploit that sparsity without specialised hardware or sparse formats.

c. Because the pruned model must be retrained from random initialisation.

d. Because only bias terms are eligible for removal.

Correct Answer: b

Explanation:

Unstructured pruning zeroes individual weights wherever they happen to be small, so the matrix keeps its original shape and the zeros sit at arbitrary positions. A standard dense matrix multiply still performs the same number of operations. Memory can be saved with a sparse storage format, but throughput usually improves only with structured patterns, such as removing whole heads, channels or the 2:4 pattern that some accelerators support directly.

(a) describes structured pruning, which is the approach that does translate into speedups.

QUESTION 5: [1 mark]

What distinguishes quantization-aware training (QAT) from post-training quantization (PTQ)?

a. QAT simulates the quantisation error during training so the weights adapt to it, while PTQ quantises an already trained model using a small calibration set.

b. QAT is applied only to activations and PTQ only to weights.

c. PTQ requires more compute than QAT because it retrains every layer.

d. PTQ can only be applied to models below one billion parameters.

Correct Answer: a

Explanation:

In QAT, fake quantisation nodes are inserted in the forward pass so the loss reflects the rounding error, letting the weights move to values that survive quantisation. The cost is a full training or fine-tuning run.

PTQ skips that: a few hundred representative examples are pushed through the model to observe activation ranges, and those statistics set the scale and zero-point of each tensor. It is far cheaper, which is why it is the default for very large models, and it usually loses a little more accuracy at low bit widths.

QUESTION 6: [1 mark]

In QLoRA, what does double quantization actually quantise a second time?

a. The attention weights, taken down to 1 bit in a second pass.

b. The gradients, before they are accumulated into the optimiser state.

c. The quantisation constants themselves, that is the per-block scale factors produced by the first quantisation.

d. Portions of the model that are reverted to FP16 whenever accuracy drops.

Correct Answer: c

Explanation:

Block-wise quantisation stores one scale factor per block of weights. With small blocks, those constants become a real fraction of the memory footprint. Double quantisation quantises the constants too, recovering roughly half a bit per parameter on average.

It is a pure memory optimisation on top of the 4-bit NF4 representation and does not touch the adapters, which stay in higher precision throughout.

QUESTION 7: [1 mark]

Which of the following statements about knowledge distillation for language models are correct?

a. Word-level distillation trains the student to match the teacher's output distribution at each token position.

b. Sequence-level distillation trains the student on sequences generated by the teacher rather than on per-token distributions alone.

c. The two can be combined so the student learns both local token probabilities and global sequence structure.

d. Sequence-level distillation requires the teacher to emit a chain-of-thought explanation for every example.

Correct Answers: a, b, c

Explanation:

(a) True: the usual objective is the KL divergence between teacher and student distributions at every position, which transfers far more information than the one-hot target alone.

(b) True: sequence-level distillation approximates the sequence distribution by decoding from the teacher and training the student on those outputs as if they were gold data.

(c) True: combining both is common and lets the student pick up token-level calibration as well as the teacher's preferred sequence structure.

(d) False: chain-of-thought distillation is one particular variant, not a requirement of sequence-level distillation.

QUESTION 8: [1 mark]

In mechanistic interpretability, what is meant by a circuit?

a. A pipeline that collects and batches training examples for an autoregressive model

b. A recurrent module inserted into a Transformer to give it additional memory

c. A subgraph of the network - particular heads, neurons and connections - hypothesised to implement a specific behaviour

d. An external tool used to adjust parameters after training has finished

Correct Answer: c

Explanation:

A circuit is a claim about mechanism: that some identifiable set of components, connected through the residual stream, is doing a particular computation. The claim is then tested by intervening on those components, for instance by ablating a head and checking that the behaviour disappears while unrelated behaviours survive.

QUESTION 9: [1 mark]

In-context copying of the pattern [A][B] ... [A] -> [B] is attributed to induction heads. Which description matches the mechanism?

a. A single head that attends to punctuation to segment the sequence into repeated units

b. A previous-token head that writes information about the preceding token, composed with a later head that attends to the token following the earlier occurrence of A and copies it

c. A feed-forward sub-layer that raises the probability of any token seen earlier in the context

d. A head that masks future positions so that repeated tokens cannot be attended to

Correct Answer: b

Explanation:

The mechanism needs two layers. A head in the earlier layer copies information about token A into the residual stream at the position of B, so that position is effectively tagged as following A. A head in the later layer, at the second occurrence of A, matches that tag, attends to the position holding B, and copies B into the output.

This composition is the standard example of how a behaviour can require two heads working together, and the sharp rise in induction head strength is what coincides with the in-context learning phase change observed during training.

QUESTION 10: [1 mark]

In the transformer circuits framing, what do the QK and OV circuits respectively determine for an attention head?

a. QK determines where the head attends; OV determines what is read from the attended position and written to the output.

b. QK determines what is written to the output; OV determines where the head attends.

c. Both determine how positional information is encoded.

d. Both are normalisation terms that keep the attention scores in a fixed range.

Correct Answer: a

Explanation:

The QK circuit, W_Q^T W_K, produces the attention pattern: it decides, for each destination position, which source positions are read. The OV circuit, W_O W_V, decides what effect the attended source position has on the residual stream at the destination.

Splitting a head this way is useful because the two can be studied separately. An induction head, for example, has a QK circuit that matches the previous-occurrence pattern and an OV circuit that acts close to a copy operation.

Option (d) confuses these with the 1/sqrt(d_k) scaling, and converting the final residual vector into vocabulary logits is the job of the unembedding matrix, not the OV circuit.
